

Section 1 — Noise Identification

Each image was visually inspected to determine the likely noise type based on its spatial characteristics.

Image	Noise Type	Justification
camera	Gaussian	Smooth, continuous fine-grained variation uniformly distributed across all intensities. No isolated outliers.
coins	Salt-and-Pepper	Scattered isolated bright and dark pixels against a dark background. Classic impulsive outlier pattern.
text	Salt-and-Pepper	Dense random white/black speckle overlaid on handwriting. Extreme pixel values with no spatial correlation.
rice	Speckle	Granular multiplicative noise; brighter grain regions are more severely affected than the dark background.
astronaut	Gaussian	Fine uniform random perturbation across all RGB channels. No impulsive outliers or multiplicative structure.

Section 2 — Method Selection

Method 1: Spatial Domain — Median Filtering

Median filtering replaces each pixel with the median value of its local neighborhood. It was selected because:

- Robust to outliers: The median statistics ignore extreme salt-and-pepper pixels, making it the most effective single method against impulsive noise.
- Edge-preserving: Unlike mean or Gaussian filters, it selects an actual pixel value from the neighborhood, preserving sharp boundaries.
- General-purpose: Also reduces Gaussian and speckle noise effectively, making it suitable across all noise types in this dataset.

Method 2: Frequency Domain — Gaussian Low-Pass Filter

Noise typically occupies high-frequency components of the Fourier spectrum. The Gaussian LPF attenuates these while preserving low-frequency structure. Implementation steps:

- Step 1 — Apply 2D Fourier Transform (`np.fft.fft2`) to convert to frequency domain.
- Step 2 — Shift zero-frequency to centre (`np.fft.fftshift`).
- Step 3 — Construct Gaussian filter: $H(u,v) = \exp(-(u^2+v^2) / 2\sigma^2)$, where σ is the cutoff parameter.
- Step 4 — Multiply spectrum by H element-wise to suppress high frequencies.
- Step 5 — Inverse shift and apply IFFT (`np.fft.ifft2`). Take real part, clip to $[0, 255]$.

Section 3 — PSNR and SSIM Explanation

PSNR — Peak Signal-to-Noise Ratio

PSNR measures the ratio of the maximum possible pixel value to the Mean Squared Error (MSE) between the clean and denoised image:

$$\text{PSNR} = 10 \times \log_{10} (255^2 / \text{MSE})$$

- Higher PSNR → lower MSE → less pixel-level distortion → better quality.
- Values above 30 dB are generally considered good; above 40 dB is excellent.
- PSNR is easy to compute but does not fully model human visual perception.

SSIM — Structural Similarity Index

SSIM compares luminance, contrast, and structural patterns between two images. Values range from 0 to 1.

- Higher SSIM (closer to 1.0) → greater perceptual and structural similarity → better denoising.
- More aligned with human visual perception than PSNR alone.
- Both metrics together give a comprehensive evaluation of denoising quality.

Section 4 — Results Table

Three parameter values were tested per method; the best (by SSIM) is reported.

Spatial: kernel sizes 3×3, 5×5, 11×11

Frequency: cutoff σ = 15, 40, 70

PART 5 — RESULTS TABLE				
Image	Method	Best Parameters	PSNR (dB)	SSIM
camera	Spatial	median, kernel=7	25.48	0.6442
camera	Frequency	Gaussian LPF, cutoff=40	25.04	0.6879
coins	Spatial	median, kernel=3	28.07	0.8536
coins	Frequency	Gaussian LPF, cutoff=40	23.38	0.5748
text	Spatial	median, kernel=3	32.01	0.9043
text	Frequency	Gaussian LPF, cutoff=15	24.78	0.6298
rice	Spatial	median, kernel=7	23.34	0.5659
rice	Frequency	Gaussian LPF, cutoff=40	24.14	0.6168
astronaut	Spatial	median, kernel=7	24.40	0.6803
astronaut	Frequency	Gaussian LPF, cutoff=40	21.93	0.6064
AVERAGE	Spatial		26.66	0.7297
AVERAGE	Frequency		23.85	0.6231

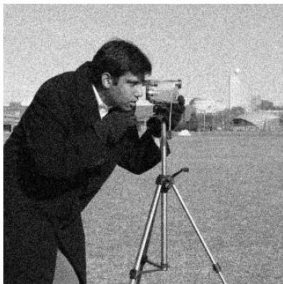
Section 5 — Visual Results

Image: camera

Clean (Reference)



Noisy Input



Spatial — Median k=7
PSNR=25.48 dB SSIM=0.6442



Frequency — Gaussian cutoff=40
PSNR=25.04 dB SSIM=0.6879

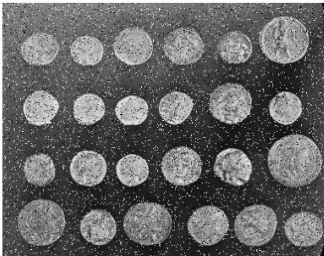


Image: coins

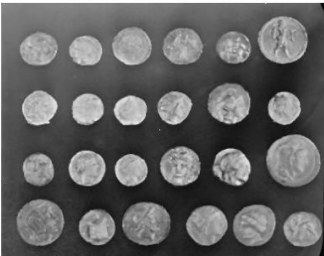
Clean (Reference)



Noisy Input



Spatial — Median k=3
PSNR=28.07 dB SSIM=0.8536



Frequency — Gaussian cutoff=40
PSNR=23.38 dB SSIM=0.5748

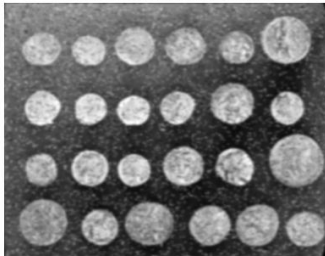
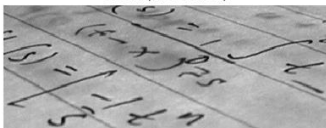
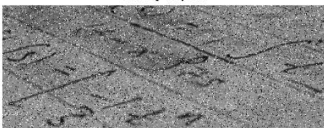


Image: text

Clean (Reference)



Noisy Input



Spatial — Median k=3
PSNR=32.01 dB SSIM=0.9043



Frequency — Gaussian cutoff=20
PSNR=25.59 dB SSIM=0.6524

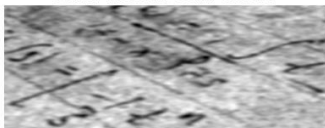
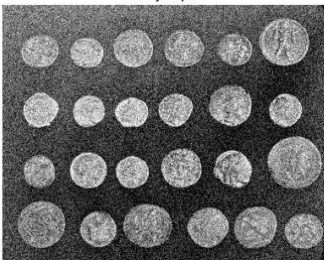


Image: rice

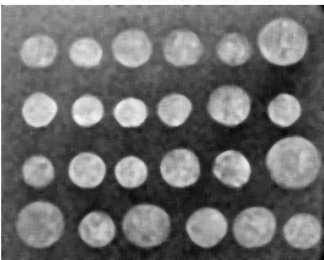
Clean (Reference)



Noisy Input



Spatial — Median k=7
PSNR=23.34 dB SSIM=0.5659



Frequency — Gaussian cutoff=40
PSNR=24.14 dB SSIM=0.6168

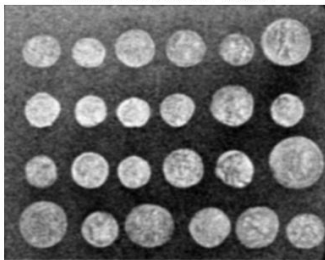


Image: astronaut

Clean (Reference)



Noisy Input



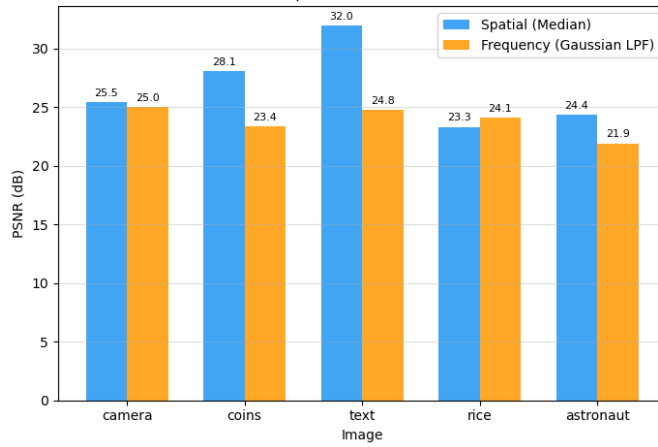
Spatial — Median $k=7$
PSNR=24.40 dB SSIM=0.6803



Frequency — Gaussian cutoff=40
PSNR=21.93 dB SSIM=0.6064



PSNR Comparison — Best Parameters



SSIM Comparison — Best Parameters

